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Teddyne LeCroy introduced the first 100GHz real-time oscilloscope, and Keysight oscilloscopes offer bandwidth up to 63GHz. "But, you need to be very careful while taking measurements at high bandwidths, even if it is 16GHz or 20GHz, as the microwave effects from cables, connectors and other peripherals tend to distort signals," informs Sanchit Bhatia, application engineer, Keysight Technologies India Pvt Ltd. The response waveform might not be absolutely clean. "One possible method to obtain a noise-free response would be to first make measurements on a network analyser and then use the response of the cable and de-embed that in the oscilloscope," adds Bhatia. [De-embedding is an attempt to make the port as ideal as possible, similar to a calibration of a radio frequency (RF) measurement system with probes or connectors used for measurement].

A precision cable, or precision probe, can be used in order to get rid of the anomalies caused by a regular cable or probe, and hence bypass the process of using an analyser. Bhatia says, "In our oscilloscopes, we have a built-in simple, mini network-analyser feature, which measures the response of the cable, or probe, by generating a high-speed pulse, and the insertion loss of the cable is measured up to double bandwidth of the scope." He adds, "The measurement performed by the scope is used to compensate for the loss with calculations done by the software. This unique feature can be used to meas-

ure the insertion loss in the time domain, which is ultimately converted into frequency domain."

Use software packages for performing different kinds of analyses

Naresh Narasimhan, country marketing manager, Tektronix India, says, "Engineers are increasingly customising their general-purpose oscilloscopes via software packages that are now available with many oscilloscopes in this class." For instance, software applications, such as various serial decode packages, vector signal analysis (VSA) software, power application and offline viewing and analysis software, allow users to customise and use their general-purpose oscilloscopes in a very specific manner.

"SignalVu VSA software, from Tektronix, that runs on the company's various oscilloscopes series, has been enabling engineers to easily characterise and validate wideband and microwave spectral events," says Narasimhan. He adds, "SignalVu-PC VSA software helps one to easily validate wideband designs."

The latest generation of oscilloscopes are also coming with Windows 7 embedded operating system (OS) platform today. Bhatia says, "With Windows 7 embedded OS comes the advantage of writing advanced software, thereby providing high analysis capabilities, such as jitter analysis or mathematical functions. New user interfaces are also powering Windows based scopes, which help users customise by cre-



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